MILITARY SPECIFICATION

CIRCUIT CARD ASSEMBLY: 10548125

(COUNTER A, RANGE RATE)

1. SCOPE

1.1 This specification covers the detail requirements for one type of circuit card assembly. The requirements of this specification together with the general requirements of MIL-C-50652 comprise the complete requirements for the range rate counter A used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on the date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-C-50652	Circuit Card Assemblies, General Specification for

DRAWINGS

U.S. Army, Frankford Arsenal

F10548125	Counter A, Range Rate

PACKING DATA SHEET

10548125	Packaging of Range Rate Counter A

(Copies of specifications, drawings, and packaging data sheets required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

FSC 1285

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3. REQUIREMENTS

3.1 Fabrication. - The assembly shall be manufactured in accordance with Drawing F10548125 and drawings pertaining thereto.

3.2 General specifications. - The assembly shall meet the require ments of MIL-C-50652.

3.3 Environmental service conditions. - As specified in MIL-C-50652.

3.4 Electrical. - Each conductor shall be electrically continuous as specified on the schematic diagram.

3.5 Performance. - For the requirements of 3.5.1 through 3.5.3.4 the following conditions shall apply.

a. Logic 0: 0.0 ±0.5 volt direct current (Vdc) level.

b. Logic 1: +3.5 ±1.0 Vdc level.

c. Positive pulse: a change in direct current (dc) level from 0.0 ±0.5 volt (V) to +3.5 ±1.0 V and return to 0 V.

d. Negative pulse: a change in dc level from +3.5 ±1.0 V to 0.0 ±0.5 V and return to +3.5 V.

3.5.1 Input power. - The input power applied between terminals P1-30, 31 and 32 (+5 V) and terminals P1-24, 25 and 26 (+5 V return) shall be 5.0 ±0.3 Vdc, at 300 milliamperes (mA), regulated, with an alternating current (ac) ripple less than 50 millivolts (mV) peak-to-peak.

3.5.2 Input signal. - The input signal shall be a positive pulse of kilo- hertz (kHz) and 0.8 ±0.1 microsecond (μs) duration.

3.5.3 Output signals.

3.5.3.1 Up count signals. - With the input logic levels applied as specified in Table I, Step 1, the output signals shall be symmetrical square waves for the set and reset side, with frequencies as specified in Table II. Comparing two consecutive set or reset stage output signals, they shall be in phase, indicating a count up direction control.

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TABLE I. - INPUT LOGIC LEVELS

	INPUT TERMINALS				
STEP	16	53	11	14	51
1	1	0	1	1	1
2	0	1	1	1	1
3	1	0	0	1	1

TABLE II. - OUTPUT SIGNALS

SET SIDE TERMINALS (TP)		RESET SIDE TERMINALS (TP)		SIGNAL FREQUENCY
7	1	13	21	10 kHz
9	2	12	20	5 kHz
56	3	10	19	2.5 kHz
37	4	21	18	1.25 kHz
33	5	19	17	625 Hz
35	6	23	16	312.5 Hz
38	7	36	15	156 Hz
40	8	22	14	78 Hz
48	9	45	13	39 Hz

3.5.3.2 Down count signals. - With the input logic levels applied as specified in Table I, Step 2, the output signals shall be symmetrical square wave for the set and reset side, with frequencies as specified in Table II. Comparing two consecutive set or reset stage output signals, they shall have a phase shift, indicating a count down direction control.

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3.5.3.3 Reset signals. - With the input logic levels applied as specified in Table I, Step 3, the output signals shall be logic levels as specified in Table III.

TABLE III. - RESET OUTPUT SIGNALS

LOGIC 1 SET SIDE TERMINALS	(TP)	LOGIC 0 RESET SIDE TERMINALS	(TP)
7	1	13	21
9	2	12	20
56	3	10	19
37	4	21	18
33	5	19	17
35	6	23	16
38	7	36	15
40	8	22	14
48	9	45	13

3.5.3.4 Pulse output signals. - With the input logic levels applied sequentially as specified in Table IV, the pulse output signals shall be as specified in Table IV.

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TABLE IV. - PULSE OUTPUT SIGNALS

STEP	INPUT TERMINALS 16	53	11	14	51	TERMINAL	OUTPUT (TP)	SIGNAL CHARACTERISTIC 1/
1	0	1	1	1	1	5	22	Positive pulse, 100 μs PRF
2	0	1	0	1	1	5	22	Logic 0 level
3	0	1	1	1	1	1	23	Positive pulse, 200 μs PRF
4	0	1	1	1	1	3	24	Positive pulse, 0.4 ms PRF
5	0	1	1	1	1	6	25	Positive pulse, 0.8 ms PRF
6	0	1	1	1	1	55	26	Positive pulse, 1.6 ms PRF
7	0	1	1	1	1	43	10	Negative pulse, 27.6 ms PRF
8	1	0	1	1	1	43	10	Logic 1 level
9	0	1	1	1	1	49	11	Positive pulse, 27.6 ms PRF.
10	0	1	1	1	1	34	12	Logic 1 level
11	1	0	1	1	1	34	12	Negative pulse, 27.6 ms PRF
12	0	1	1	0	0	49	11	Positive pulse, 100 μs PRF

1/ All pulses shall have a duration of 0.8 ±0.1 μs.

4. QUALITY ASSURANCE PROVISIONS

4.1 General. - The quality assurance provisions of MIL-C-50652 shall apply to the assembly covered by this detail specification.

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4.2 First article inspection. - As specified in MIL-C-50652.

4.3 Inspection provisions. - As specified in MIL-C-50652.

4.4 Special sampling. - As specified in MIL-C-50652.

4.5 Inspection equipment. - As specified in MIL-C-50652 and 4.5.3

4.5.1 Government-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.2 Contractor-furnished inspection equipment. - As specified in MIL-C-50652.

4.5.3 Test equipment. - In conjunction with 4.5 the following test equipment shown in Table V shall be utilized in the performance of the applicable tests as specified in 4.6.3.

TABLE V. - TEST EQUIPMENT LIST

ITEM NUMBER	DESCRIPTION
1. Power supply	5.0 ±0.3 Vdc, regulated. Ac ripple: less than 50 mV peak-to-peak. Capacity: 300 mA.
2. Pulse generator	20 kHz positive pulse; 3.0 ±1.0 Vdc amplitude, 0.8 ±0.1 μs wide. Pulse rise and fall times: less than 0.1 μs.
3. Oscilloscope	Dual channel; rise time of scope plus measuring probe 10 ns or less. Sweep period: 50 ns per division or greater. Input resistance: greater than 8 megohms. Input capacitance: less than 25 pF.
4. Test fixture	As specified in Figure 1. See Table VI for test fixture parts list.

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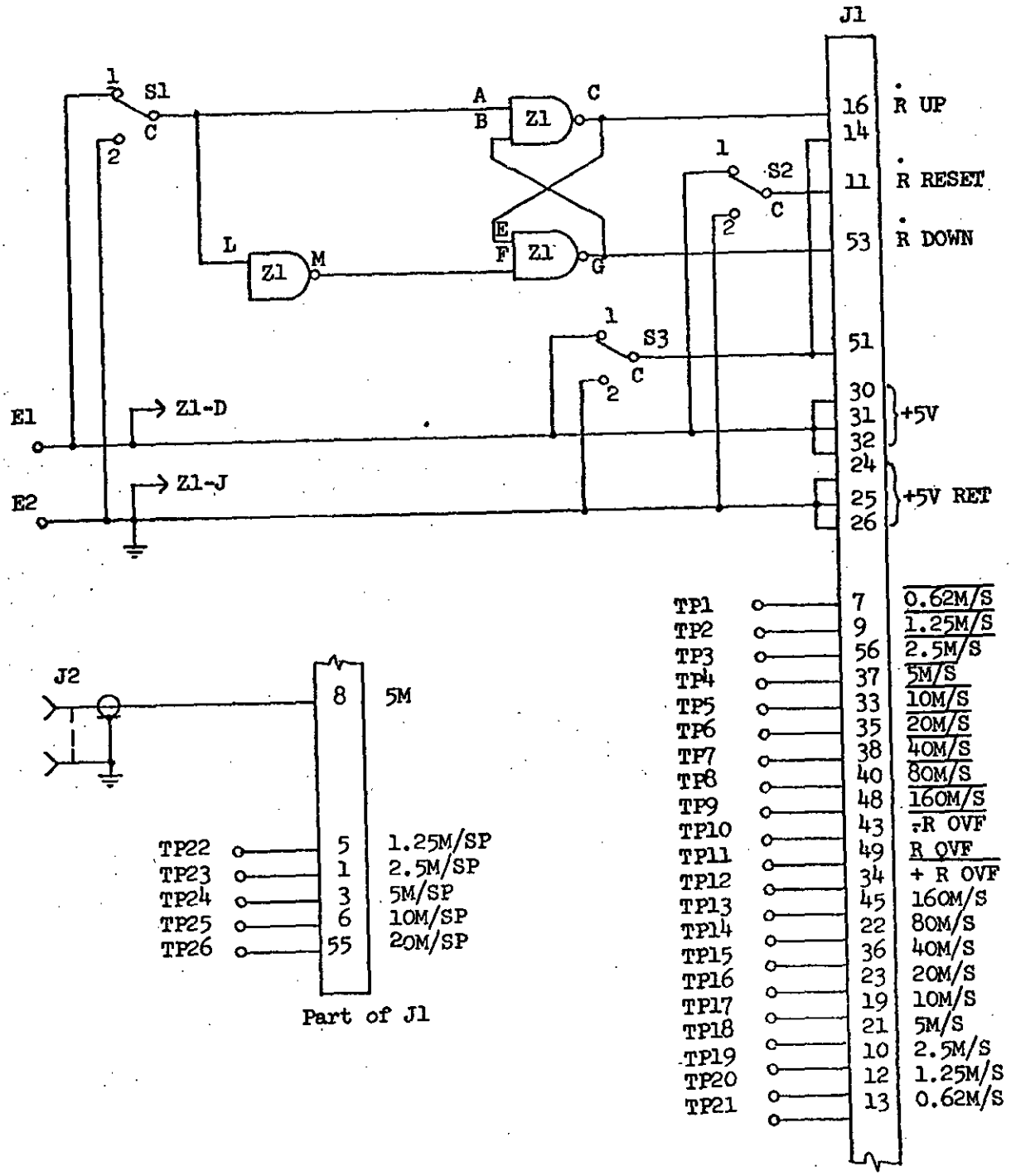


FIGURE 1. TEST FIXTURE SETUP.

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TABLE VI. - TEST FIXTURE PARTS LIST

DESIGNATION	NOMENCLATURE AND DESCRIPTION
E1	Binding post, H. H. Smith 257-R or equivalent.
E2	Binding post, H. H. Smith 257-B or equivalent.
J1	Connector; any of the following or equivalent: (1) 5893232-1, Malco Mfg. Co., Inc., Chicago, Illinois. (2) 500946-01, National Connector Corp., Minneapolis, Minn. (3) 07-5422-0198-01, Elco Corp., Willow Grove, Pa.
J2	Connector, receptacle, UG-25B/U or equivalent.
S1	Switch SPDT.
S2	Same as S1.
S3	Same as S1.
TP1 thru TP26	Tip jack, H. H. Smith 1501 or equivalent.
Z1	Integrated circuit, SG-140-03 Sylvania or equivalent.

4.6 Test methods and procedures.

4.6.1 Environmental. - As specified in MIL-C-50652.

4.6.2 Electrical. - Continuity shall be inspected by standard measuring equipment (SME) and visual examination to determine compliance with 3.4.

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4.6.3 Performance. - The selected range registers shall be tested as follows: Connect range register A and test equipment, items 1 through 4, as shown in Figure 2. Pulses and logic levels shall be as specified in 3.5. Use switches S1 through S3 to apply logic levels as follows:

a. S1 - In position 1 applies logic 1 to P1-53 and logic 0 to P1-16, in position 2 applies logic 0 to P1-53 and logic 0 to P1-16.

b. S2 - In position 1 applies logic 1 to P1-11 and in position 2 applies logic 0 to P1-11.

c. S3 - In position 1 applies logic 1 to P1-14 and 51, in position 2 applies logic 0 to P1-14 and 51.

Use oscilloscope, item 3, to observe input and output signals as specified in 4.6.3.2 through 4.6.3.3.4.

4.6.3.1 Input power. - Using power supply, item 1, apply input power as specified 3.5.1.

4.6.3.2 Input signal. - Using pulse generator, item 2, set up and ap apply input signal, as specified in 3.5.2, to J2 on test fixture, item 4.

4.6.3.3 Output signals.

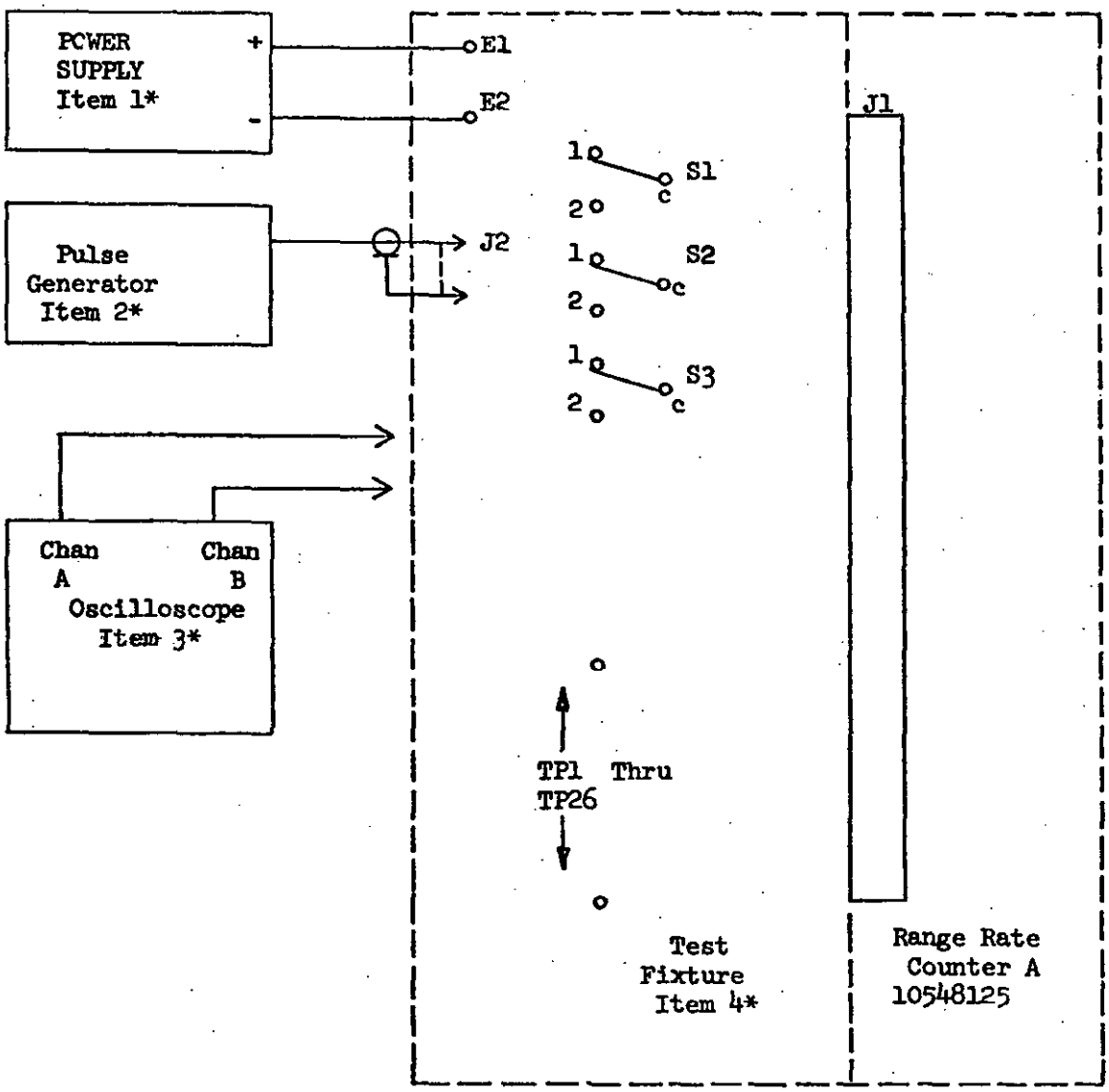
4.6.3.3.1 Up count signals. - Using switches S1 through S3 apply logic levels as specified in Table I, Step 1, and observe the output signals as specified in Table II. Use oscilloscope channels A and B to compare set and reset output stages for phase as specified in 3.5.3.1.

4.6.3.3.2 Down count signals. - Using switches S1 through S3 apply logic levels as specified in Table I, Step 2, and observe the output signals as specified in Table II. Use oscilloscope channels A and B to compare set and reset output stages for phase shift as specified in 3.5.3.2.

4.6.3.3.3 Reset signals. - Using switches S1 through S3 apply logic levels as specified in Table I, Step 3, and observe the output levels as specified in Table III.

4.6.3.3.4 Pulse output signals. - Using switches S1 through S3 apply logic levels as specified in Table IV and observe the output signals specified in Table IV.

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*Refer to 4.5.3

FIGURE 2. RANGE RATE COUNTER A TEST SETUP.

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5. PREPARATION FOR DELIVERY

5.1 Packaging, packing, and marking. - Packaging, packing, and marking shall be in accordance with Packaging Data Sheet 10548125. The level of protection shall be as specified in the procurement document.

6. NOTES

6.1 Intended use. - The range rate counter A is used in Range Computer CP-888/VPS-2, part of Radar Set AN/VPS-2. The range rate counter A is used to accumulate a scaled acceleration input for derivation of range rate information.

6.2 Ordering data. - Procurement documents should specify the following:

a. Title, number, and date of this specification.

b. Applicable packaging data sheet number (see 5.1).

c. Selection of applicable level of preservation packaging and packing.

d. Level of protection required.

e. Applicable stock number.

f. Requirement for first article submission.

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Preparing Activity:

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Project No. 1285-A246

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 22-R255.

INSTRUCTIONS: This sheet is to be filled out by personnel, either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for, obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity. Comments and suggestions submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or serve to amend contractual requirements.

SPECIFICATION

ORGANIZATION

CITY AND STATE

CONTRACT NUMBER

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO *(If "yes", in what way?)*

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity - Optional)*

DATE

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